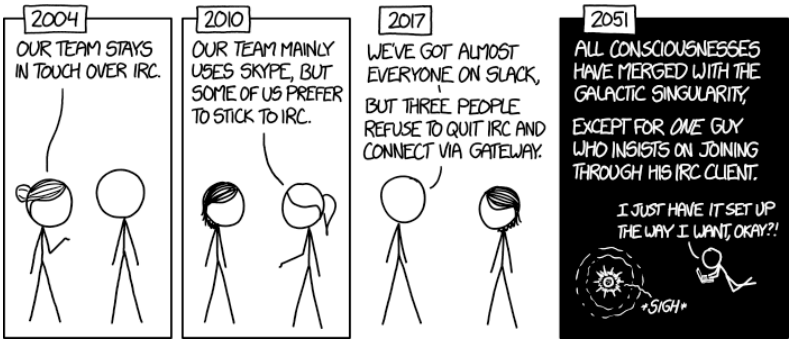
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**Course 22051: Signals and Linear Systems in Discrete Time**

22051

# Project assignment - Space travel




source: xkcd.com

Deliverable:
Code, data, and presentation video of max 5 minutes duration presenting your design choices and justifications. Pick code snippets, present relevant figures, prepare explanatory sound examples, ..., be creative!

Deadline:
Last week of the semester.

Mission:
Develop your competences and have fun!

Bastian Epp: 45 25 39 53 (office), e-mail: bepp@dtu.dk

## Your entrance ticket to the "final assessment"

In order to be included the final assessment, I will ask you to work during this semester on the implementation of a larger project. This could be an example of you working in a start-up where you need to provide a solution to a customer. And you need to be able to back-track everything to the very beginning in case there need to be modifications.

The basic idea is that you won't be able to do much in the beginning of the course (How should you? You just started on DSP!). But as we progress, you will be able to start implementing certain elements - and continuously think about how to make the individual elements play together. The individual assignments that each of you will be working on align fully with the group project - so time spent on the individual project is time spent on the group project! It will be essential that you work together as a team - and that you work on this in a continuous manner. There might occur the need for ongoing changes and modifications - so don't consider your code to be finished before it's done!

A key element in this project will be that you share knowledge. And you are not only encouraged to share knowledge within your own group, but I also highly encourage you to share knowledge **across groups**! Learn from each other how these problems can be tackled - and then find your own way to implement them! **Share knowledge and insights, not code!** You need to submit your own code and a detailed description of all the elements in your system as a group at the beginning of week 13. Copying code or text is plagiarism and can not be accepted. But please go ahead and chat!

The presentation of your solution will happen through a video of maximum duration 5 minutes. Besides a walktrough of the project, provide the following elements - they will be factors in the final evaluation:

1. Select `<Number_of_group_members>` elements of the system and justify your design choices. Explain (or demonstrate) the consequence of a "wrong" / "unsuitable" design choice for that element of the system. ("We did ...because of....if that was violated, the consequence would be...").

2. Name for each of the above elements at least one alternative and what the impact of that would be on the overall project.

3. Any items you could not provide (if any), where you got stuck, and how you tried to solve the problem.

# Babelsberg called - need some sound

It happened! We started our first company. Yes - it is in a garage. But we have pizza (vegetarian), beer and are full of enthusiasm and knowledge! You've been hired as the lead DSP engineers for an upcoming sci-fi film called "Andromeda-Bohr-Pauli Zone". Your task is to create a **complete 40-60 second audio track** that simulates spaceship radio communication experiencing various space phenomena.

## The plot and the outcome

You are flying with a space ship <NAME> through a pretty dull part of the galaxy. Super boring. In an anomaly of the Andromeda-Bohr-Pauli zone (known to reflect all electro-magnetic radiation) your screen linked to the outside camera shows the mirror image of your space craft and the graphical representation of your ships' name[1] on the hull appearing on your screen. Looks cool! You think back to the fun times of your DSP class where you nailed that problem.

To distract you, you switch on your interstellar radio station where you can hear a piece of music. Odd what they consider music these days - it is just some weirdo synthesizer-let-me-do-whatever sound put together[2]. Just for the sake of it, you look at a spectrogram of the sound coming in - it might provide some hidden code (who knows). You try to apply a number of different filters to see how the signal changes - at least with one of them you manage to remove that narrowband noise signal of the anomaly known to cover 6-8 kHz.

Damn! You forgot all about that instrument you won from that weirdo in that dodgy space bar. That dude really believed that you can change spectral resolution by increasing the sampling frequency! You got to know that it simulates some classical instrument from earth. You un-dust your instrument-detector[3], hit the atomic string and watch the display: It says:

> ''The instrument you are playing belongs to the group of <YOU FIGURE IT OUT>''.

OK - time to tune into the news. You switch channels and listen to a news feed. Nothing exciting. But when passing the orbit of the Chebychev-moon, you start picking up a broadband noise that makes your signal pretty noisy. Well. Nothing you can do - and you having some band-limited radio equipment doesn't help. Even worse, You are about to cross through the shadow of that moon known for its high contents of metal - so you will likely be cut off the signal transmission for about 2 seconds or so.

On your trajectory approaching and leaving the moon shadow, you hear that the radio transmission is automatically cut into half bandwidth and reduced quantization (to help your error-correction algorithm). You start getting some first drop-outs while entering the moon shadow. Behind the moon, you are left with some few seconds of no signal transmission before the signal step-wise starts to return to normal until you can enjoy the news again. Great - time to wrap up this mission and to move on. All that is left is to do the video-logging that you'll have to provide before getting your certificate.

[1] See individual assignment "Sound is art"
[2] See individual assignment "Add/multiply/synthesize"
[3] See individual assignment "Tune in"

### What the audience should hear:

Given that the visuals are recorded elsewhere, you'll need to prepare the audio and send that back to the studio. There is some slack on the exact timing - we'll get that fixed. The total duration of the sound clip will be between 40 and 60 seconds and should contain the following elements:

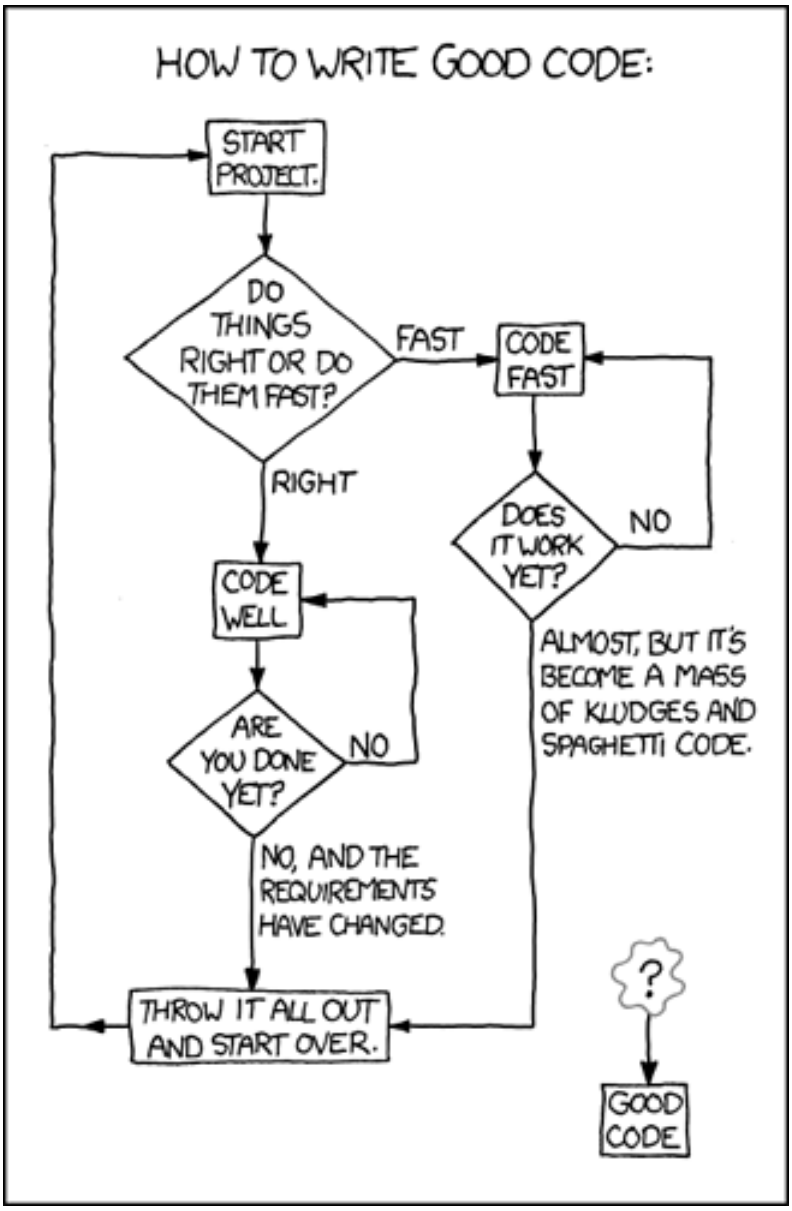
- Calm transmission at the beginning (clear synthesizer music, optional: space ship background sounds)
- Instrument sound with detection procedure (recorded sound, optional: device sound)
- Gradual interference entering an anomaly zone (narrowband noise 6-8 kHz)
- Sudden signal degradation near a metallic moon (broadband noise + drop-outs)
- Temporary blackout crossing behind the moon ( 2 seconds full signal loss)
- Gradual recovery emerging from the moon shadow
- Return to normal communication at the end

## How to get it done and what to submit

If you need support, then please reach out to your network (the other groups) and post questions etc. on the communication channel. Feel free to post code on DTU LEARN - and see how others solve the problem. You will find your own way of implementing it - and this is what we will be discussing during the next weeks. You are also welcome to bring up some specifics during the weekely small-group discussions where we can tackle things specifically.

The format of the submission will be a .zip file with the naming convention `GRP_#_group_assignment.zip` (with `#` your group number) containing the final video and all the code and data required to solve the assignment. I recommend using GIT for code development - you might provide a link to the REPO together with your submission. You can check your solution against the corresponding rubric on LEARN which will be used for assessment.

Good luck!




source: xkcd.com

## Acknowledgements

This project assignment is the result of teamwork - with essential contributions from enthusiastic students that are interested in improving teaching for the generations to come. Thanks to MBR, XH, IÞ for the key ideas!

# A Appendix

## A.1 Core requirements, mandatory components, deliverables, suggested time line, and learning objectives

This is both a technical demonstration and a professional audio production that needs to sound convincing while rigorously implementing all DSP concepts from this course. Below you find some tools that might help you to organize your work. All these are just suggestions and you are free to organize your work in the way it fits you best. In Table A.1 you see a suggestion for how to break the assignment into elements. The empty column can be used to brainstorm about how to connect to relevant concepts from the course.

Table A.1: Core requirements

| **Scene element** | **Audio effect** | **DSP technique required** |
|---|---|---|
| Camera feed | Visualized ship name | |
| Opening transmission | Clear radio broadcast | |
| Instrument detection | Brief moment analyzing equipment (with acoustic feedback?) | |
| Anomaly interference | Narrowband hum at 6–8 kHz | |
| Radiation storm | Crackle/hissing noise overlay | |
| Moon shadow entry | Signal cuts progressively | |
| Complete blackout | $\sim$ seconds of silence | |
| Signal return | Gradual restoration to clarity | |

## A.2 Mandatory technical components

Table A.2 list the components that need to be in the final product. All concepts will be covered througout the course. In case you need a concept that has not yet been covered or that you just find exciting, feel free to browse the course website, consult the book, and provide the relevant references.

Table A.2: Mandatory technical components

| **Requirement** | **Specification** | **Check** |
|---|---|---|
| Filter types | At least one recursive (IIR) and one non-recursive (FIR) filter implemented | |
| Bilinear transform | Analog-to-digital filter conversion via bilinear transform with documented pre-warping calculations (in code) | |
| Power spectral density (PSD) | Estimation using Welch's method or periodogram with parameter justification | |
| Pole-zero analysis | z-plane pole-zero plot for at least one recursive filter with stability discussion | |
| Spectrogram | Visualization showing signal at relevant scene phases | |
| Sampling frequency | Justification referencing Nyquist criterion for the chosen sampling rate | |
| Frequency resolution | Discussion linking window length to DFT properties and spectral resolution trade-offs | |
| Terminology | Proper English DSP terminology used consistently throughout documentation | |
| Literature citations | Minimum 2 external sources cited for filter design choices and theoretical foundations | |

## A.3 Group-internal deliverables

A suggestion for project deliverables is provided in Table A.3. The weightings provide an idea of how much each of them will be filling in your overall workload.

Table A.3: Group-internal deliverables

| **Item** | **Description** | **Format** | **Weight** |
|---|---|---|---|
| Main pipeline notebook | Complete workflow from raw signal to final audio track with step-by-step documentation | MATLAB Live-Script or Jupyter Notebook | 35% |
| Filter and effect modules | Documented recursive/non-recursive filters + noise generators with parameter explanations | Separate .m/.py files or clearly demarcated sections | 25% |
| Final audio track | 40-60 second stereo WAV file ready for movie use, all scene phases integrated | .wav file (44.1 kHz or higher, 16+ bit) | 20% |
| Technical analysis report | Document explaining design choices, theory connections, and trade-offs | PDF document and code comments | 10% |
| Individual contributions | Self-assessment rubric completion + video preparation for assigned module | Forms + video (5 min per group) | 10% |

## Project time line

Finally, Table A.4 presents a suggested time line that should provide you with ample time to implement the project after having covered the basics of the course.

Table A.4: Project timeline: Compressed 8-week schedule (weeks 6–13)

| **Weeks** | **Milestone** | **Deliverable** | **Check** |
|---|---|---|---|
| 6 | Project kickoff + signal pipeline | Scenario assignment + module allocation + clean audio loading | |
| 7 | Anomaly interference + filtering | Band-stop filter design docs + pre/post filtering spectrograms | |
| 8 | Radiation storm + PSD analysis | Broadband noise generator + PSD estimates (Welch/periodogram) | |
| 9 | Moon shadow + temporal analysis | Windowing functions + attenuation curves + pole-zero plots | |
| 10 | Signal recovery | Dropout simulation + recovery crossfade | |
| 11 | Integration + soundtrack draft | Full soundtrack draft + technical report outline + self-assessment | |
| 12 | Final audio mastering + report | Complete audio track + code documentation + video draft | |
| 13 | Final submission | Complete .zip package + 5-min video presentation | |

## Learning objectives

Table A.5 provides you an overview over which learning objectives are covered by which part of the project.

Table A.5: Group assignment tasks mapped to learning objectives

| **Task** | **Specific Activity** | **LO** | **Evidence Required** | **Verification Method** |
|---|---|---|---|---|
| Opening Transmission (Week 6) | Record/clean audio loading + sampling rate selection | LO1 | Nyquist justification document | Peer assessment Q&A |
| Opening Transmission (Week 6) | Anti-aliasing filter design before resampling | LO9 | Minimum viable rate calculation | Written documentation (code) |
| Anomaly Interference (Week 7) | FIR band-stop filter design (6–8 kHz) | LO3 | Coefficient calculation + frequency response plot | Code inspection |
| Anomaly Interference (Week 7) | IIR band-stop via bilinear transform + pre-warping | LO4 | Pre-warping calculations + analog prototype docs | Technical review |
| Radiation Storm (Week 8) | Broadband noise generation + spectral shaping | LO6 | Noise model description + PSD comparison | Spectrogram analysis |
| Radiation Storm (Week 8) | PSD estimation using Welch's method or periodogram | LO5, LO6 | Parameter justification + window size trade-off | PSD plot with labels |
| Moon Shadow (Week 9) | Windowing functions for temporal masking | LO7 | Window type justification (Hann/Hamming/Blackman) | Frequency domain plots |
| Moon Shadow (Week 9) | Pole-zero plot for recursive filter stability | LO7 | Stability proof + z-plane visualization | Peer assessment Q&A |
| Complete Blackout (Week 10) | Signal interpolation during dropout | LO3 | Interpolation algorithm (linear/spline/cubic) | Code implementation |
| Complete Blackout (Week 10) | Quantization depth reduction simulation | LO1, LO9 | Bit-depth vs. SNR analysis | Audio comparison file |
| Signal Return (Week 11) | Cross-fade transitions between scenes | LO7, LO10 | Smooth envelope design + click prevention | Final audio track |
| Signal Return (Week 11) | Resampling effects after bandwidth reduction | LO2 | FFT vs. DTFT vs. DFT explanation | Written documentation (code) |
| Instrument Detection (Week 12) | Harmonic analysis + classification demo | LO5 | Spectral centroid/formant tracking | Classification accuracy metric |
| Integration (Week 12) | Final soundtrack assembly (60–90s WAV) | LO10 | Complete pipeline reproducibility | Code runs without errors |
| Documentation (Week 13) | Technical documentation (comments or report) with DSP terminology | LO11 | Consistent English terminology audit | Code review |
| Documentation (Week 13) | Literature citations for filter design choices | LO12 | Minimum 2 sources properly cited | Bibliography check |
| Oral Defense (Week 13) | 5-minute Q&A on assigned module | LO1–LO12 | Ability to explain design rationale | Overall assessment |